Adwait V. Datar

Postdoctoral Researcher

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Research profile

My research spans information geometry, control theory, and optimization, with an increasing interest in learning and control of dynamical systems. My current work concerns natural-gradient optimization and Wasserstein-geometric approaches to divergence and regularization. I also investigate how policy architectures affect the performance and robustness of embodied agents whose policies are learned through reinforcement learning, with a longer-term view toward robotic applications.

Experience

Postdoctoral Researcher — Jul 2023-Present
Institute for Data Science Foundations, TU Hamburg — *Hamburg, Germany*

- Develop theory at the interface of information geometry and learning, including natural-gradient optimization and Wasserstein-geometric regularization for learning and control.
- Co-author of work published at *ICML*, *JMLR*, *TMLR*, and *SIAM Journal on Scientific Computing*; teach and supervise across machine learning, information geometry, mathematics for machine learning.

Doctoral Researcher + Postdoctoral Researcher — Nov 2016-Jun 2023
Institute of Control Systems, TU Hamburg — *Hamburg, Germany*

- Developed modular methods for distributed control of heterogeneous multi-vehicle systems, using robust-control tools and integral quadratic constraints to analyze uncertain cooperative dynamics; doctorate defended in November 2022.
- Coauthor of work published at *IEEE TAC*, *ACC*, *ECC*, and *CDC*;
- Planned and supervised the development of an indoor multi-quadrotor laboratory, coordinating equipment selection, hardware setup, and student contributions; guided the implementation of distributed source-seeking algorithms.
- Contributed technical material and preliminary results to the successful Phase 2 proposal for DFG Priority Programme 1914 on cyber-physical networking.

Research Engineer — Jul 2012-Aug 2014
Mercedes-Benz R&D India / Daimler R&D — *Pune, India / Stuttgart, Germany*

Education

Dr. rer. nat. (PhD) — 2022
TU Hamburg — *Hamburg, Germany*
Distributed Control of Multi-Vehicle Systems Using Modular Architectures.
Supervisor and first examiner: Prof. Herbert Werner.
Other examiners: Prof. Peter J. Seiler and Prof. Marko Lindner.

MSc, Mathematical Modelling in Engineering (MathMods) — 2016
University of L'Aquila, University of Hamburg, Gdańsk University of Technology — *Europe*
Joint Erasmus Mundus programme; thesis at the University of Stuttgart.

BE, Mechanical Engineering — 2012
University of Pune — *Pune, India*

Teaching and Supervision

- Currently delivering tutorials for **Mathematics for Machine Learning** (winter 2026–27); previously delivered tutorials for **Information Geometry** (summer 2026, 2025), **Machine Learning I** (summer 2025, 2024, 2023) and **Machine Learning II** (winter 2024–25, 2023–24)
- Delivered **Machine Learning II** lectures and theoretical tutorials in winter 2025–26, using materials developed jointly with Nihat Ay and Manfred Eppe in earlier cohorts.
- Delivered tutorials for **Optimal and Robust Control** (2018–2023), **System Identification** (2022, 2019, 2018), and **Control Systems Theory and Design** (2018); produced reusable and recorded teaching material.
- Supervised or co-supervised **20 bachelor's and master's students**, including four ongoing theses, alongside student projects; six supervised projects led to peer-reviewed publications in control theory, data-driven control, and robotics.
- Advised doctoral researchers informally on research questions, proof development, and theoretical analysis, including work leading to joint publications.
- Delivered an industry tutorial on integral quadratic constraints for IAV GmbH.

Selected recent research outputs

1. V. Stein, **A. Datar**, and N. Ay. "Well-Posed KL-Regularized Control via Wasserstein and Kalman-Wasserstein KL Divergences." *Proceedings of the 43rd International Conference on Machine Learning*, PMLR 306:115711-115746, 2026. [link]
2. **A. Datar** and N. Ay. "Information Geometry for Wasserstein KL Divergence of Gaussian Measures on $\mathbb{R}^n$." Under review at *Information Geometry*; arXiv:2503.24022, v3, 2026. [link]
3. **A. Datar** and N. Ay. "Wasserstein KL-Divergence for Gaussian Distributions." *Geometric Science of Information*, 2025, pp. 91-101. [link]
4. **A. Datar** and N. Ay. "Convergence Properties of Natural Gradient Descent for Minimizing KL Divergence." *TMLR*, 2025. [link]
5. N. Ay, J. van Oostrum, and **A. Datar**. "On the Natural Gradient of the Evidence Lower Bound." *JMLR* 26, 2025. [link]
6. C. Datar, **A. Datar**, F. Dietrich, and W. Schilders. "Systematic Construction of Continuous-Time Neural Networks for Linear Dynamical Systems." *SIAM Journal on Scientific Computing* 47(4), 2025. [link]
7. **A. Datar**, C. Hespe, and H. Werner. "Robust Performance Analysis of Cooperative Control Dynamics via Integral Quadratic Constraints." *IEEE Transactions on Automatic Control* 70(1), 2025. [link]

Communication and open research

- Presented natural-gradient research at FDIG 2025, University of Tokyo, and at TU Hamburg's Mathematics of Machine Learning 2025 conference.
- Contributed to WiMAS, open-source wireless multi-agent simulation software, and produced Tutorial recordings and experimental robotics demonstrations.

Tools and languages

Tools: Python, MATLAB, LaTeX.

AI-assisted work: Claude Code and OpenAI Codex to support research and teaching activities.

Languages: English (C1), German (B1), Marathi (native).